

RAJKUMAR S

Lead Engineer | Backend and Integration Architecture | Java, Kafka, AWS

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PROFESSIONAL SUMMARY

Lead Engineer with 4.5 years building enterprise B2B SaaS, leading 3 engineers and owning 5 core CMMS modules serving clients across 4 regions. Backend-focused full stack engineer: distributed systems, Apache Kafka event-driven architecture, multi-tenant SaaS on AWS, Vue.js, React, and React Native. Forward Deployed Engineer building custom AI-agent solutions and Technical Account Engineer for enterprise accounts; scheduling systems spanning 100,000+ assets.

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL

Frontend and Mobile: Vue.js, React, React Native

Backend, Messaging, and Cloud: REST API design, GraphQL, microservices, Apache Kafka, event-driven architecture, message queues, asynchronous processing, AWS (EC2, S3), Docker

Databases: MySQL, relational data modeling, query optimization, indexing, performance tuning

AI and Automation: AI agents, LLM integration, custom AI skills and tooling, workflow automation

Architecture and Practices: Distributed systems, system design, multi-tenant SaaS, offline-first synchronization, role-based access control, third-party integrations, Agile, Scrum, code review, Git, CI/CD

Leadership and Client-Facing: Team leadership, mentoring, technical account management, forward deployed engineering, stakeholder management, client workshops

PROFESSIONAL EXPERIENCE

Facilio | Enterprise facilities and maintenance management SaaS (CMMS) | Chennai, India

Lead Engineer, CMMS Modules and Forward Deployed Engineer | Jun 2025 - Present

- Lead 3 engineers owning 5 core CMMS modules (Work Order, Planned Maintenance, Job Plans, Inventory, Procurement) for enterprise clients across 4 regions.
- Own quarterly roadmap, architecture, code quality, release readiness, and production stability for the module group end to end.
- Re-architected Planned Maintenance V3 as product owner and lead developer onto event-driven Apache Kafka processing, scheduling maintenance across 100,000+ assets with idempotent, fault-isolated generation.
- Resolved a platform-wide access-control limit after the permission bitmask exhausted all 63 usable bits, designing a 64-bit permission action registry and authoring the reference architecture adopted organization-wide.
- Built custom AI-agent solutions for enterprise clients as Forward Deployed Engineer, automating client workflows and cutting manual effort.
- Mentor 3 engineers through design and code review; distributed reusable AI skills and architecture documentation adopted team-wide.

Full Stack Developer and Technical Account Engineer | Apr 2022 - May 2025

- Delivered 6 enterprise modules end to end (Work Order, Planned Maintenance, Inventory, Procurement, Workplace, Visitor) across Java and MySQL backend, Vue.js and React web, and React Native mobile.
- Built the SFG-20 compliance integration with bidirectional GraphQL sync and write-back, adopted across approximately 10 European enterprise accounts.
- Managed escalations, requirements, and solution design for key enterprise accounts across 3 regions as Technical Account Engineer, running client workshops with the SFG-20 standards body.
- Simplified workflows and interfaces by analyzing observed customer usage patterns, reducing steps and support load for client teams.
- Optimized high-volume Work Order list and detail queries, substantially reducing page load times on large enterprise datasets.
- Built offline-first React Native synchronization so field technicians create and update work orders without network connectivity.

KEY PROJECTS

Planned Maintenance V3 Scheduling Engine | Java, Apache Kafka, MySQL, Vue.js

- Built a configurable scheduling engine with 2 planner types (time-based and task-based), complex recurrence rules, seasonal triggers, multi-site execution, and duplicate prevention across 100,000+ assets.

- Migrated work order generation from synchronous request processing to queue-backed asynchronous workers with idempotent processing, per-record failure isolation, and job-status tracking.
- Designed reusable PM templates with override detection and dissociation tracking so template changes propagate without overwriting customer-level customization.
- Delivered the full Vue.js creation and summary experience, bulk import pipeline with pre-scan validation and dedupe, nightly forward scheduler, and a V5 API surface for external automation.

Platform Access-Control Re-architecture | Java, MySQL

- Root-caused a platform-wide failure to add new permissions: the enforcement bitmask had consumed all 63 usable bits of a signed long, and a prior migration had been abandoned half-finished with a write path nothing read from.
- Designed and shipped a Permission Action Registry with a second 64-bit permission space, extending the enforcement path, role model, and package import/export without touching the legacy bit space.
- Authored the reference architecture covering 3 historical permission schemes and the live request-to-decision enforcement path, now the organization-wide standard before any access-control change.

SFG-20 Compliance Integration | Java, GraphQL, REST APIs

- Built a bidirectional integration with the Facilities-iQ GraphQL API: incremental change-based sync, schedule and version reconciliation, job-plan diffing against locally customized plans, and write-back of task completions.
- Delivered end to end in direct collaboration with the SFG-20 standards body and European clients; adopted across approximately 10 European enterprise accounts as their compliance backbone.

Work Order Management System | Java, Vue.js, React, React Native, MySQL

- Generalized the Work Order module into a unified system flow across web, mobile, and offline, delivering consistent behavior for the platform's largest and most customized module.
- Built the React Native offline-first changes with data synchronization so field technicians create and update work orders without connectivity.
- Replaced hard-coded per-customer logic with a configurable action framework: role-based button visibility, per-role permissions, and status-transition configuration.

Risk-Based Maintenance | Java, Vue.js, MySQL

- Built configurable risk management into maintenance: risk matrix setup, category and impact definitions, priority rating ranges, and color-coded matrix visualization.
- Extended risk to the task level with classification and risk category on job plans, PM tasks, and work-order tasks, wired through create, import, API, and export paths.

Inventory, Procurement, and Field Service Commercials | Java, MySQL

- Owned the procurement lifecycle across 5 document types (Purchase Requests, RFQs, Quotes, Purchase Orders, Invoices) and built the Invoice module from scratch with validation, multi-step approval chains, and financial reconciliation.
- Integrated procurement with Work Order and Budget for end-to-end cost traceability, and delivered client, vendor, and service contracts with automated invoice and credit-note generation on work-order completion.

Workplace and Visitor Management | Vue.js, Mapbox, AutoCAD, Java

- Architected a CAD-to-web workflow rendering AutoCAD drawings as interactive Mapbox floor plans for desk reservation and space utilization reporting.
- Built visitor check-in workflows with multi-step approval chains and badge printing automation.

EDUCATION

Bachelor of Engineering, Computer Science and Engineering | Sri Krishna College of Technology, Coimbatore | 2018 - 2022